E-COMMERCE SALES ANALYSIS USING MICROSOFT EXCEL

Introduction

This project focuses on performing E-Commerce Sales Analysis using Microsoft Excel. The main objective of this project is to clean, transform, analyze, and visualize e-commerce sales data to generate meaningful business insights.

The project includes Customer, Product, Store, and Sales datasets, which were cleaned and organized to improve data quality and prepare the data for analysis. Various Excel techniques such as data cleaning, transformation, descriptive statistics, pivot tables, charts, and dashboard creation were used throughout the project.

This project demonstrates how Microsoft Excel can be effectively used for business reporting and decision-making in real-world environments.

Objectives of the Project

The key objectives of this project are:

• To clean and prepare raw datasets for analysis

• To identify and handle missing values

• To remove duplicate records and improve data accuracy

• To transform and organize datasets for reporting

• To perform descriptive statistical analysis

• To analyze data using Pivot Tables

• To create charts for better data visualization

• To build an interactive dashboard for business insights

Dataset Description

This project includes four datasets related to e-commerce business operations.

Customer Dataset

Contains customer-related details such as:

• Customer ID
• Name
• Age
• Gender
• City
• State
• Loyalty Level

Product Dataset

Contains product-related details such as:

• Product ID
• Product Name
• Category

• Brand

• Price

• Stock

Store Dataset

Contains store-related information such as:

• Store ID

• Store Name

• Region

• City

• Store Type

Sales Dataset

Contains sales transaction details such as:

• Sales ID

• Order Date

• Customer ID

• Product ID

• Store ID

• Quantity

• Payment Type

• Discount

• Total Amount

These datasets were cleaned and prepared for meaningful business analysis

Data Cleaning and Transformation Process

Data cleaning and transformation were performed to improve dataset quality and consistency.

Handling Missing Values

Tasks Performed:

• Identified missing values

• Filled blank records where necessary

• Standardized incomplete data

Removing Duplicate Records

Tasks Performed:

• Checked duplicate rows

• Removed duplicate records

Correcting Inconsistent Data

Tasks Performed:

• Corrected formatting inconsistencies

• Fixed spacing and text-related issues

• Standardized names and categories

Data Transformation

Tasks Performed:

• Split and organized columns

• Applied sorting and filtering

• Converted data into structured table format

Formatting Techniques

Tasks Performed:

• Applied Currency Formatting

• Applied Percentage Formatting

• Formatted dates properly

• Applied Conditional Formatting

Descriptive Statistics Analysis

Descriptive statistical analysis was performed to summarize and understand dataset behavior.

The following statistical measures were analyzed:

• Mean

• Median

• Mode

• Minimum Value

• Maximum Value

• Standard Deviation

• Count

Purpose:

To understand data patterns and summarize business information effectively.

Pivot Table Analysis

Pivot Tables were created to analyze different business areas.

Customer Analysis

• Customer distribution by state

• Loyalty level analysis

Product Analysis

• Category-wise product analysis

• Brand and stock analysis

Sales Analysis

• Payment type analysis

• Sales and revenue analysis

Store Analysis

• Region-wise and city-wise store analysis

Purpose:

To identify trends and generate meaningful business insights.

Charts and Data Visualization

Charts were created to visually represent analytical insights.

Charts Used:

• Bar Charts

• Column Charts

• Pie Charts

• Line Charts

Purpose:

• To simplify data interpretation

• To identify patterns and trends

• To improve reporting quality

Dashboard Creation

An interactive dashboard was created in Microsoft Excel using:

• Pivot Charts

• Slicers

• Filters

Dashboard Features:

• Customer insights

• Payment analysis

• Product performance

• Store performance

Purpose:

To provide an easy and interactive view of business performance.

Tools and Features Used

The following Microsoft Excel features were used:

• Data Cleaning

• Remove Duplicates

• Find and Replace

• Sorting and Filtering

- Conditional Formatting
- Pivot Tables
- Pivot Charts
- Slicers
- Descriptive Statistics ToolPak
- Dashboard Design

Conclusion

This project successfully demonstrated the process of cleaning, transforming, analyzing, and visualizing e-commerce sales data using Microsoft Excel. Through descriptive statistics, Pivot Tables, charts, and dashboard creation, meaningful insights were generated regarding customers, products, stores, and sales performance.

This project improved practical knowledge in Excel-based data analysis and business reporting.

Acknowledgment

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